

AI ARCHITECT

From Fundamentals to Production-Ready Systems



01 THE ARCHITECT PROGRAMME

Where most courses *stop,*

*we begin. A syllabus
on the four production
layers that separate a
demo from a
deployment.*

*The Architect Programme teaches the
four layers that decide whether your work
survives a **board review**, an **enterprise
security audit**, and the morning after
deploy.*

— PROGRAMME CHARTER, ED. II

This is the engineer's deep-dive. You already call
an API and run a basic retrieval. The Architect
Programme covers advanced retrieval,
evaluation, integration, and the engineering of trust.
Seventy per cent hands-on labs; thirty per cent
architecture; nothing else.

16:00
HOURS

IV.
PILLARS

70/30
LAB/LECTURE

L. II
ARCHITECT

THE FOUR PILLARS

I. Advanced RAG
INPUT LAYER · 4 HRS

II. Evals & Benchmarks
QUALITY LAYER · 4 HRS

III. MCP Engineering
INTEGRATION LAYER · 4 HRS

IV. Trust & Production
OPERATIONS LAYER · 4 HRS

AN AXIOM OF THE PROGRAMME

*"From your first day, you are never stuck for
more than an hour."*

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Nine sections. Four pillars. One capstone. The arc of the Architect Programme set out in order of ascent.

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I. SECTION ONE

Four production layers.

Skills enterprises pay \$200K+ to hire for.

The Architect Programme is organised as four interlocking layers — Input, Quality, Integration, and Operations. Each is taught in four hours of structured instruction with hands-on labs. Together they answer the questions every enterprise AI evaluation eventually asks.

I.

Advanced RAG

INPUT LAYER · 4 HRS

Lift retrieval quality 15–30%

Hybrid search + cross-encoder reranking; top-50 → top-5 funnels that beat naïve RAG.

Reason across documents

Agentic RAG: decompose · validate · iterate. GraphRAG for multi-hop entity logic.

Route smart, not slow

Adaptive RAG — simple to naïve, complex to agentic. Cut cost 5x without quality loss.

Personalise with memory

User-scoped retrieval with Mem0, LangMem, Letta. Memory that compounds, not resets.

III.

MCP Engineering

INTEGRATION LAYER · 4 HRS

Build the layer others consume

Tools · Resources · Prompts. JSON-RPC 2.0. Schemas that guide LLM behaviour.

Enterprise-grade auth

OAuth 2.1 as Resource Server. RFC 8707 audience scoping. Auth0 / Okta integration.

Production-grade resilience

Circuit breakers, structured logging, audit trails, cost tracking, supply-chain hardening.

Scale stateless

Horizontal scaling, well-known discovery, rolling updates without breaking agents.

II.

Evals & Benchmarks

QUALITY LAYER · 4 HRS

Stop shipping on vibes

RAGAS metrics — faithfulness, context precision, answer relevance, drift detection.

Score agents, not just answers

DeepEval's 50+ metrics. AgentBench. Tool-call accuracy. Trajectory scoring across 9 engines.

Calibrate LLM-as-judge

Pairwise rubrics, position-bias correction, judge selection that survives upgrades.

Block bad releases

CI eval gates that fail PRs on regression. Braintrust + Langfuse + Promptfoo pipelines.

IV.

Trust & Production

OPERATIONS LAYER · 4 HRS

Pass enterprise security review

OWASP LLM01 injection defence. Llama Guard, NeMo Guardrails, Presidio PII redaction.

Lock down RAG & MCP access

Row-level ACLs in vector stores. Multi-tenant retrieval. Audience-scoped tokens.

Real HITL, not theatre

Risk-tagged tools, approval queues, pause / resume agent runs, feedback into evals.

Ship audit-ready

Langfuse traces, cost attribution, EU AI Act tier classification, SOC2 controls, playbooks.

II. SECTION TWO

From *Essentials* to *Architect*.

Essentials gave you literacy. The *Architect* Programme gives you production. The L1 syllabus answers "what is it?"; the L2 syllabus answers "how do you ship it through enterprise security review?"

A CONSUMER OF AI

AI Essentials ^{L. I}

A foundation in fluency. The student leaves able to use the tools others have built, with clear models and shipped first projects.

- First chatbot with basic memory
- Introduction to RAG: vector + chunking
- First agent — a news analyser
- Using MCP servers as a consumer
- Prompt engineering fundamentals
- SQL Intelligence demonstration

A PRODUCER OF AI SYSTEMS

AI *Architect* ^{L. II}

A discipline of production. The student leaves able to architect, evaluate, integrate, and harden AI systems that survive board review and enterprise audit.

- Agentic RAG: decompose · route · validate · iterate
- GraphRAG, hybrid retrieval, adaptive routing
- Eval-gated CI with RAGAS + DeepEval
- Building & operating custom MCP servers
- Guardrails, ACLs, HITL approval queues
- Langfuse observability & EU AI Act compliance

AN AXIOM

"L1 teaches you to be fluent. L2 teaches you to be trusted."

L1 OUTCOME

Literacy

L2 OUTCOME

Production

III WHAT YOU WILL BUILD

Four projects.

One capstone.

A board-ready portfolio. Each pillar produces a project an enterprise would deploy; the capstone integrates all four into a single trust-hardened multi-tenant application.

I.

Multi-Source *Agentic RAG* System

PILLAR I · ADVANCED RAG

Decompose · HyDE · rewrite · parallel retrieve · merge · sufficiency-score loop with budget caps. Hybrid search + Cohere reranking + memory-aware personalisation.

II.

Eval-Gated *Release Pipeline*

PILLAR II · EVALS & BENCHMARKS

RAGAS suite + DeepEval + LLM-as-judge running in CI. Regression detection, golden datasets, dashboards in Braintrust + Langfuse.

III.

Custom *MCP Integration Layer*

PILLAR III · MCP ENGINEERING

OAuth 2.1 Confluence MCP server with audience-scoped tokens (RFC 8707), structured logging, audit trails, observability — deployed and consumed by Claude Code.

IV.

Trust-Hardened *Production App*

CAPSTONE · ALL FOUR PILLARS

Multi-tenant RAG + MCP with row-level ACLs, HITL approval queue, prompt-injection defence, eval-gated release, Langfuse traces & EU AI Act compliance dashboard.

PREREQUISITES	<i>Required</i>	<i>Technical</i>	<i>Recommended</i>
	AI ESSENTIALS OR SHIPPING EXPERIENCE	PYTHON & TYPESCRIPT; REST APIS; GIT	VECTOR DBS · OAUTH BASICS

PILLAR

I.

INPUT
LAYER

Advanced *RAG*.

From naïve vector search to multi-hop, memory-aware retrieval. Five modules; six labs; the foundation on which the rest of the programme stands.

4:00 hrs

INPUT · PILLAR I
5 MODULES · 6 LABS

Thesis. *In 2026, retrieval is the bottleneck — not generation. Hybrid search and cross-encoder reranking lift RAGAS scores 15–30%; agentic loops handle multi-hop questions that vector search alone cannot.*

i. RAG Beyond Basics

45 min

Hybrid retrieval & reranking

Vector + BM25 hybrid search, RRF fusion, cross-encoder reranking (Cohere, Voyage), top-50 → top-5 funnel, query rewriting. Where naïve RAG breaks and how to fix it.

ii. Agentic RAG: Decompose · Validate · Iterate

45 min

Lab on Kapi's production agentic-rag pipeline

Query decomposition, HyDE, sufficiency scoring (relevance · specificity · confidence), multi-hop with LLM-call budgets and deadlines. Streaming pipeline as async generator.

iii. GraphRAG & Knowledge Graphs

45 min

Multi-hop reasoning over entities

Entity & relationship extraction, knowledge graph construction, community detection. Microsoft LazyGraphRAG (0.1% indexing cost vs full GraphRAG). When graph beats vector.

iv. Adaptive RAG: Query Routing

45 min

The 2026 production pattern

Query classifier routes simple → naïve, complex → agentic, relational → graph. Cost-vs-quality dashboards. Routing rules as YAML; fallback chains; A/B comparison.

v. Personalisation & Memory in RAG

45 min

Mem0 · LangMem · Letta

Working / short-term / long-term / episodic memory. User-scoped retrieval, entity memory, session summarisation. When memory beats retrieval, when they compose.

PRACTICA · HANDS-ON LABS

Six Engineering Reps

Pillar I lab fixture: Kapi's production agentic-rag pipeline.

i. Hybrid + reranking pipeline (Pinecone + Cohere)

ii. Agentic RAG with sufficiency loop & budget caps

iii. GraphRAG with Neo4j + LazyGraphRAG

iv. Adaptive RAG router with cost dashboard

v. Memory-aware RAG with Mem0 user state

vi. RAGAS evaluation of your pipeline

PILLAR

II.

QUALITY
LAYER

Evals & *Benchmarks.*

The pillar nobody builds — and the one that decides who ships. A discipline of measurement: golden datasets, calibrated judges, and continuous-integration gates.

4:00 hrs

QUALITY · PILLAR II
5 MODULES · 6 LABS

Thesis. *The 2026 standard for engineering-led teams: a lightweight framework for CI gating (DeepEval / RAGAS) paired with a platform for traces and dashboards (Braintrust / Langfuse). This is how confidence is shipped.*

i. Why Evals Are the Pillar Nobody Builds 45 min

Failure modes & the 3-tier golden hierarchy

Four eval types: offline, online, A/B, red-team. The 3-tier golden dataset model (base · blueprint · production thumbs-downs). The "5 standard + 5 blueprint-specific" criteria pattern from Kapi's eval engine.

ii. RAG Evals with RAGAS 45 min

Faithfulness · context precision · answer relevance

RAGAS metric deep-dive: faithfulness, context precision, context recall, answer relevance. Ground-truth-free patterns. Running RAGAS in CI. What good scores actually mean — and what they hide.

iii. Agent & Tool-Use Evals 45 min

DeepEval · AgentBench · 9 evaluator engines

DeepEval's 50+ research-backed metrics with pytest. AgentBench for multi-step success. The 9 evaluator engines (llm-judge, exact-match, semantic-similarity, contains, human-review + 4 RAGAS). Tool-call accuracy & trajectory evaluation.

iv. LLM-as-Judge: Calibrate the Critic 45 min

Pairwise · rubrics · bias correction

Pairwise comparison vs absolute scoring. Rubric design that survives judge swaps. Position bias, length bias, self-preference — and how to calibrate them out. Choosing the right judge model.

v. Eval-Driven Development 45 min

CI gating · regression · eval-to-HITL bridge

CI eval gates as release thresholds. Regression detection on prompt & model changes. The eval-to-HITL bridge: failed evals automatically promote to human review queues (Kapi's hitl-bridge).

PRACTICA · HANDS-ON LABS

Six Quality Reps

Pillar II lab fixture: Kapi's eval engine — 20 YAML criteria across 9 evaluators.

i. RAGAS suite on your retrieval pipeline

ii. DeepEval pytest integration in CI

iii. LLM-as-judge with bias calibration

iv. Build a release gate (fail PR on regression)

v. Wire eval failures into a HITL review queue

vi. Run a benchmark slice (MTEB, HELM)

PILLAR

III. MCP Engineering.

INTEGRATION
LAYER

Build the integration layer — don't just consume it. Five modules; six labs; the engineering of trustworthy enterprise integration.

4:00 hrs

INTEGRATION · PILLAR III
5 MODULES · 6 LABS

Thesis. 2026's MCP is no longer "how do I expose tools?" It is "how do I run an OAuth Resource Server with audience-scoped tokens, structured logging, and stateless horizontal scaling?" That is engineering — and it is what separates demo MCP from production MCP.

i. MCP Architecture

45 min

Host · Client · Server · three primitives

JSON-RPC 2.0 foundation. Tools, Resources, Prompts. Transports: stdio (local), SSE (legacy), Streamable HTTP (2026). Host/Client/Server boundaries. The N×M problem MCP solves.

ii. Building MCP Servers

45 min

Schemas that guide LLM behaviour

Server.ts anatomy. JSON Schema design for LLM-facing APIs — descriptions that steer Claude toward correct usage. Build a Confluence MCP server from scratch in 30 minutes.

iii. OAuth 2.1 + Resource Indicators

45 min

Enterprise auth done right · RFC 8707

MCP servers as OAuth Resource Servers. PKCE flow. RFC 8707 audience-scoped tokens. Auth0 / Okta integration. Scope naming conventions. PKCE pitfalls and token leakage.

iv. Resilience & Observability

45 min

Production servers don't break silently

Error handling, retries with backoff, circuit breakers. Structured JSON logging with correlation IDs. Cost tracking for paid APIs. Distributed tracing. Audit logs. Caching strategies.

v. Deployment & Scale

45 min

Stateless · discovery · supply chain

Stateless server design for horizontal scaling. Container deployment. .well-known discovery, registry metadata. Rolling updates without breaking running agents. Supply-chain risk (Smithery Oct 2025 incident).

PRACTICA · HANDS-ON LABS

Six Integration Reps

Pillar III lab fixture: a Confluence MCP server you build, deploy, and consume from Claude Code.

i. First MCP server in 30 min (Confluence search)

ii. Schema design that guides Claude correctly

iii. Add OAuth 2.1 + PKCE to your server

iv. Wire Langfuse observability into MCP calls

v. Deploy to Container Apps with rolling updates

vi. Audit-trail every tool call

PILLAR

IV. Trust & Production.

OPERATIONS
LAYER

Guardrails · Access Control · HITL · Observability ·
Compliance. The pillar that lets your work pass enterprise
security review.

4:00 hrs

OPERATIONS · PILLAR IV
5 MODULES · 6 LABS

Thesis. Every enterprise AI evaluation asks the same three questions: "How do you do row-level security on RAG?" "What is your HITL story?" "How do you defend against prompt injection?" This pillar answers all three.

i. Guardrails & Output Safety 45 min

OWASP LLM01 · NeMo · Llama Guard · Presidio

Prompt injection defence (OWASP LLM01). Jailbreak red-teaming. NeMo Guardrails & Llama Guard for input/output filtering. Structured output enforcement (Pydantic, JSON Schema). PII detection & redaction with Microsoft Presidio.

ii. Access Controls for RAG & MCP 45 min

Row-level security in vector stores

Document-level permissions. Row-level security in pgvector / Azure AI Search / Pinecone. Multi-tenant retrieval (per-user namespaces vs prompt-time filtering). ACL injection at retrieval. MCP audience-scoped tokens.

iii. HITL Patterns 45 min

Gates at layer boundaries · risk-tagged tools

The "gate at any layer boundary" model. Tool risk tagging (read · write · destructive) drives a tool router that pauses agent runs before side effects. Approval queues, pause/resume, feedback into evals. Lab uses Kapi's HITL queue + tool router.

iv. Observability & Tracing 45 min

Langfuse · Arize Phoenix · OTEL

What to trace, what to log, what to alert on. Debugging hallucination chains across multi-step agents. Cost attribution per user / per feature. Latency flame graphs. Self-hosting vs managed.

v. Compliance & Audit 45 min

EU AI Act · SOC2 · incident response

EU AI Act tier classification (minimal · limited · high · prohibited). SOC2 Type II controls translated to AI systems. Audit logs that satisfy auditors. Data residency, retention, right-to-deletion. Incident response playbook for LLM apps.

PRACTICA · HANDS-ON LABS

Six Trust Reps

Pillar IV lab fixture: Kapi's production HITL queue + tool router.

- i. Wrap an unsafe agent with NeMo Guardrails
- ii. Add row-level ACLs to a vector store
- iii. Risk-tag tools and pause writes before dispatch
- iv. Build an approval queue with reviewer takeover
- v. Stand up Langfuse + cost dashboard
- vi. Red-team your own RAG for prompt injection

SECTION EIGHT · THE CAPSTONE

One project, *all four pillars.*

TRUST-HARDENED MULTI-TENANT APP

Adaptive RAG (Pillar I) · automated eval gate in CI (II) · custom OAuth 2.1 MCP server (III) · row-level ACLs, HITL approval queue, injection defence, Langfuse traces & an EU AI Act checklist (IV). Peer-reviewed and graded against a production-readiness rubric.

I. ADVANCED RAG Retrieval Stack HYBRID Vector + BM25 with RRF fusion RERANK Cohere · Voyage cross-encoders GRAPH Neo4j · LazyGraphRAG ADAPTIVE Query classifier & router	II. MEMORY Memory Architecture WORKING Current conversation SHORT Session context LONG Mem0 · LangMem · Letta EPISODIC User-scoped events	III. EVALS Eval & Benchmark Stack RAGAS RAG-specific metrics DEEPEVAL 50+ pytest metrics BRAINTRUST CI gating & dashboards JUDGE Calibrated rubrics
IV. MCP MCP Stack PRIMITIVE Tools · Resources · Prompts TRANSPORT Stdio · SSE · Streamable HTTP AUTH OAuth 2.1 + RFC 8707 DISCOVERY .well-known + registries	V. TRUST Trust & Guardrails INJECTION NeMo · Llama Guard PII Microsoft Presidio OUTPUT Pydantic · JSON Schema ACCESS Row-level ACL in vectors	VI. OPERATIONS Production Ops Stack TRACING Langfuse · Phoenix · OTEL HITL Approval queue · routing COMPLIANCE EU AI Act + SOC2 AUDIT Per-tool, per-user logs

Production Readiness

REQUIRED TO GRADUATE

<i>RAG.</i>	Hybrid retrieval, reranking, sufficiency check, query routing
<i>Evals.</i>	Golden dataset, RAGAS in CI, regression gate, calibrated judge
<i>MCP.</i>	OAuth 2.1, audience-scoped tokens, audit logs, structured errors
<i>Trust.</i>	Injection defence, row-level ACLs, HITL approval, PII redaction
<i>Ops.</i>	Langfuse tracing, cost dashboard, EU AI Act tier, incident playbook

IX CERTIFICATION

The Architect Certificate.

Awarded upon a 70% overall score across continuous assessment, capstone, and a technical evaluation. Graded against a production-readiness rubric reviewed by working architects.

COMPONENT		WEIGHT
Continuous Assessment	Practical labs and project milestones across the four pillars.	20%
Capstone Project	Trust-hardened multi-tenant AI application — peer reviewed against rubric.	50%
Technical Evaluation	Architecture defence, evaluation strategy, and best-practice oral.	30%
PASSING THRESHOLD		70% overall

THE PLATFORM · INCLUDED

Beyond the sixteen hours, *permanent access to the alumni platform.*

Career Coaching - Personalised AI roadmap 1-on-1 coaching sessions - Resume & portfolio review - Interview preparation - Job placement support	Networking & Community 2,500+ alumni network - Industry connections - Weekly office hours - Project showcase events - Peer study groups
Challenges & Projects - Weekly AI challenges - Leaderboard competitions - Capstone guidance - Real-world case studies - Portfolio building	Continuing Study - All future cohort access - Live workshops & clinics - New module releases - Production playbooks - Architectural reviews

CONTINUING PATHS · AFTER ARCHITECT

<i>Agentic AI</i>	Enterprise agent strategy, governance, multi-agent architectures	2 DAYS
<i>MAS · 16 Pillars</i>	Multi-agent system engineering — coordination, blackboard, topology	2 DAYS
<i>MCP Engineering</i>	Specialist path — transports, scaling, multi-agent integration	2.5 DAYS
<i>Vibe Coding</i>	AI-assisted development — Claude Code, Cursor, Copilot workflows	3 DAYS

"From your first day, you are never stuck for more than an hour."

Modern AI Pro

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